

RETRIEVING P - T STRUCTURE IN PANCHROMATIC TRANSMISSION SPECTRA

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MOTIVATION

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- We know about **increased requirements** for p - T profiles in idealised scenarios (e.g., Line+2012, Rocchetto+2016, Blečić+2017)
 - *Indirect influence of p - T profile on transmission spectra*

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- We know about **increased requirements** for p - T profiles in idealised scenarios (e.g., Line+2012, Rocchetto+2016, Blečić+2017)
- More **detailed look** into individual observations
 - *Complexity threshold? Inherent biases from over-simplified models?*
 - *Extending Schleich et al. (2024, S24) to multiple JWST and Ariel cases*

SIMULATED DATA

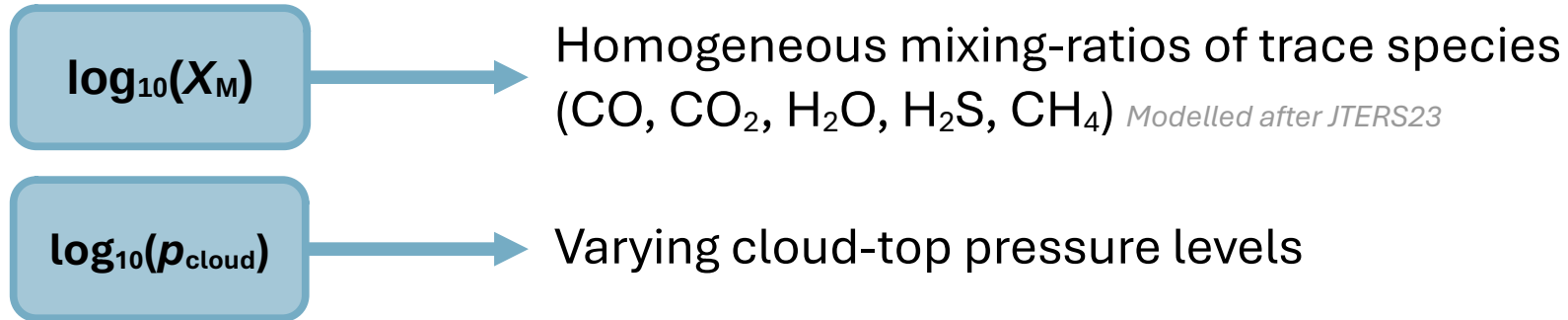
Modelled after WASP-39 b (consistent with S24)

- H_2/He – dominated atmosphere (hot Jupiter) with fixed R_p and M_p
 - *Modelled using TauREx3 (Al-Refaie et al. 2021, 2022)*

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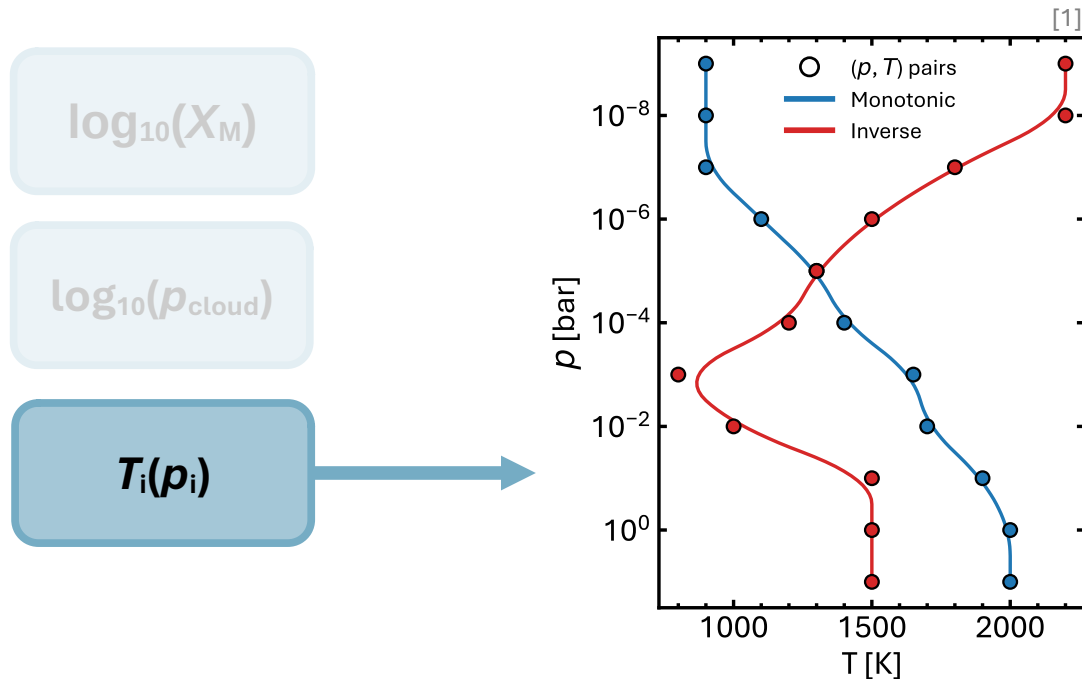
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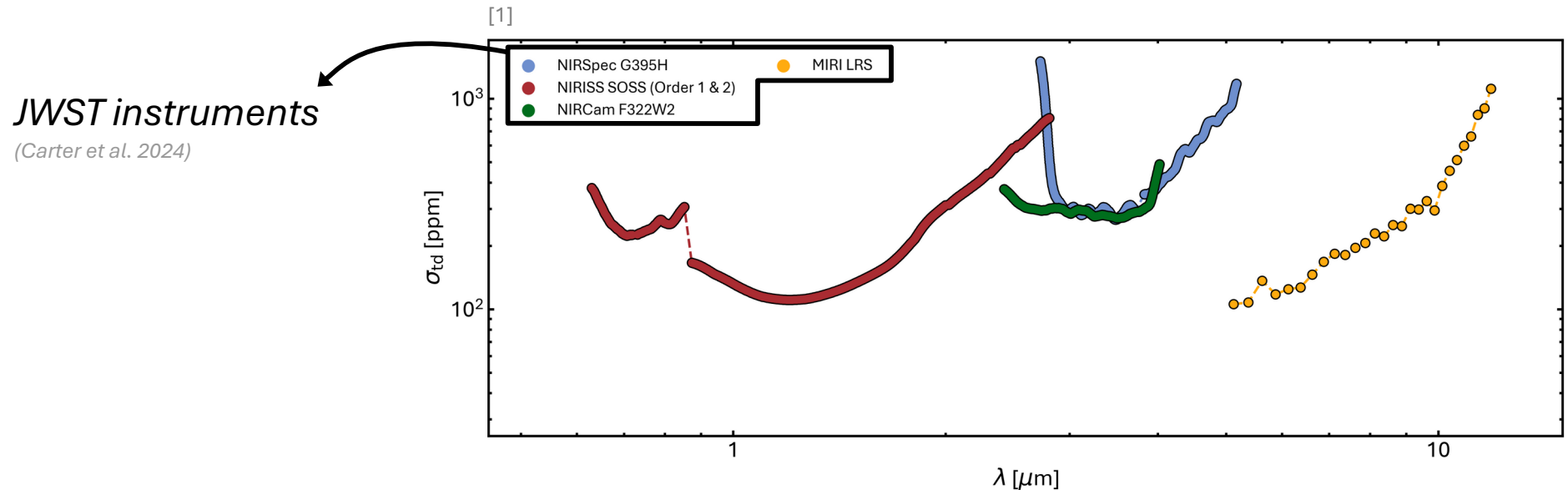
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- *Input profile = 11 points*
- *Generic profiles of two complexity levels*

SIMULATED DATA

- H_2/He – dominated atmosphere (hot Jupiter) with fixed R_p and M_p
- Synthetic spectra from realistic instrument noise



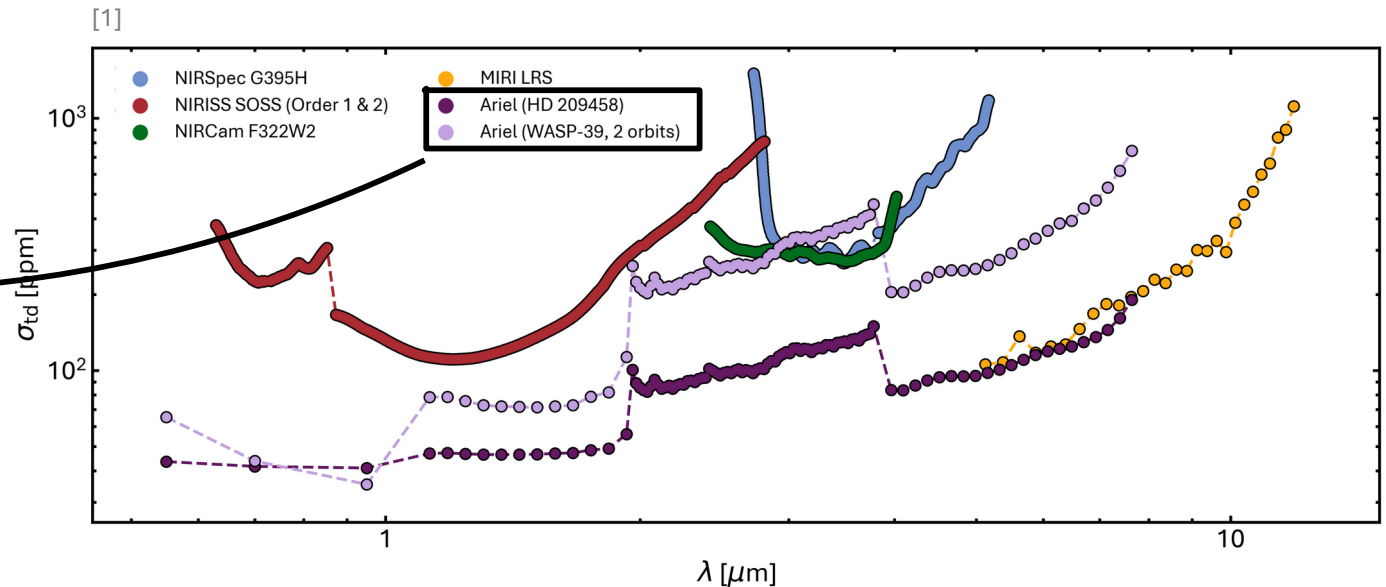
SIMULATED DATA

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JWST instruments

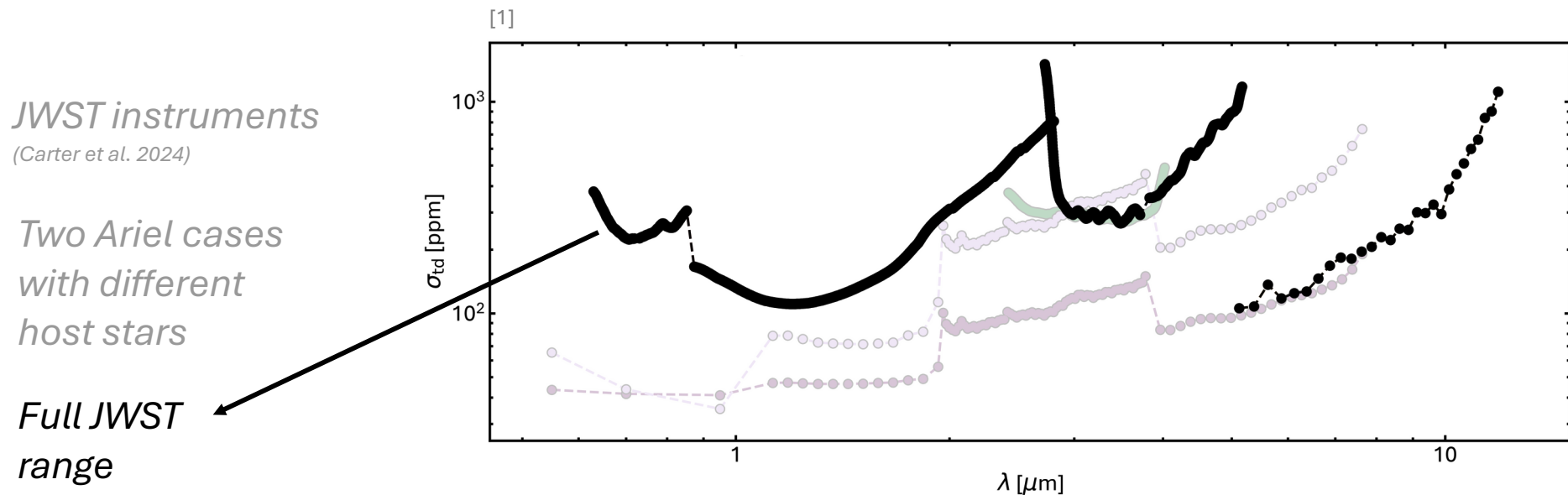
(Carter et al. 2024)

Two Ariel cases
with different
host stars



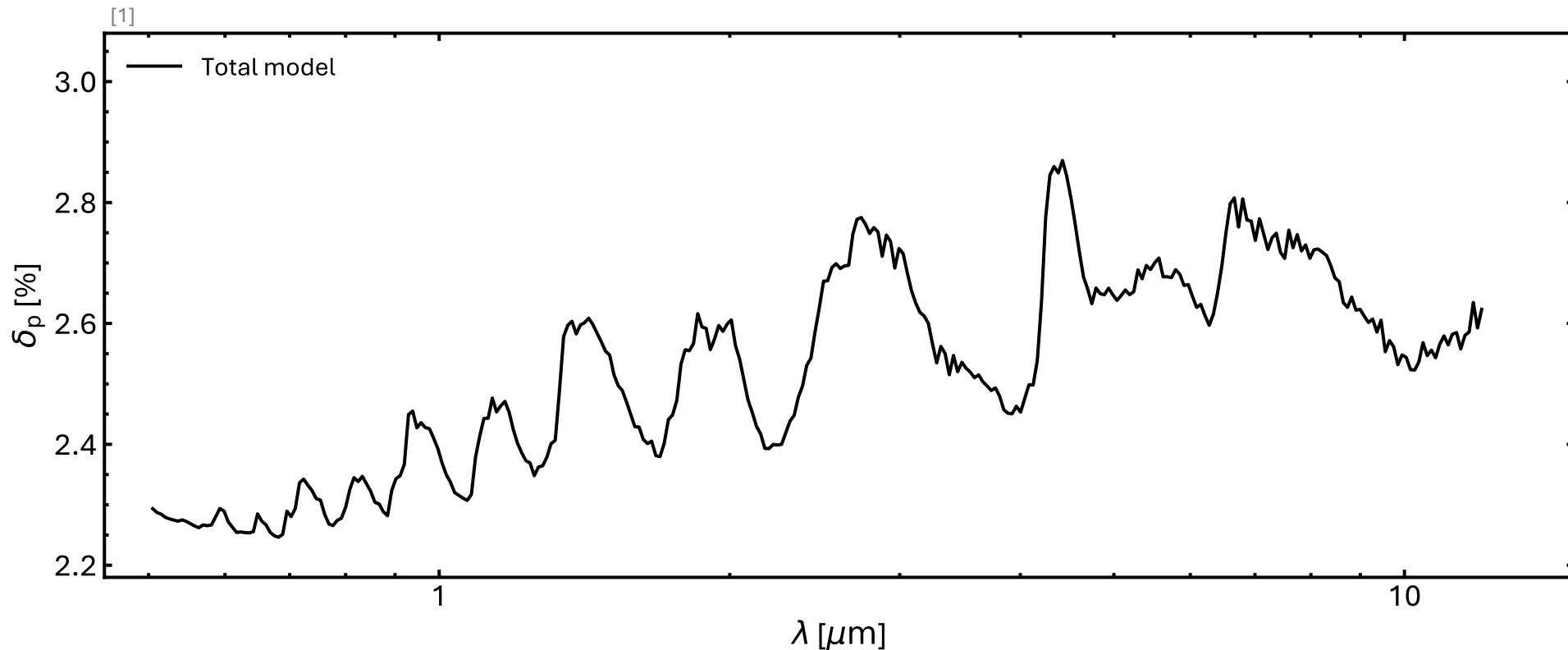
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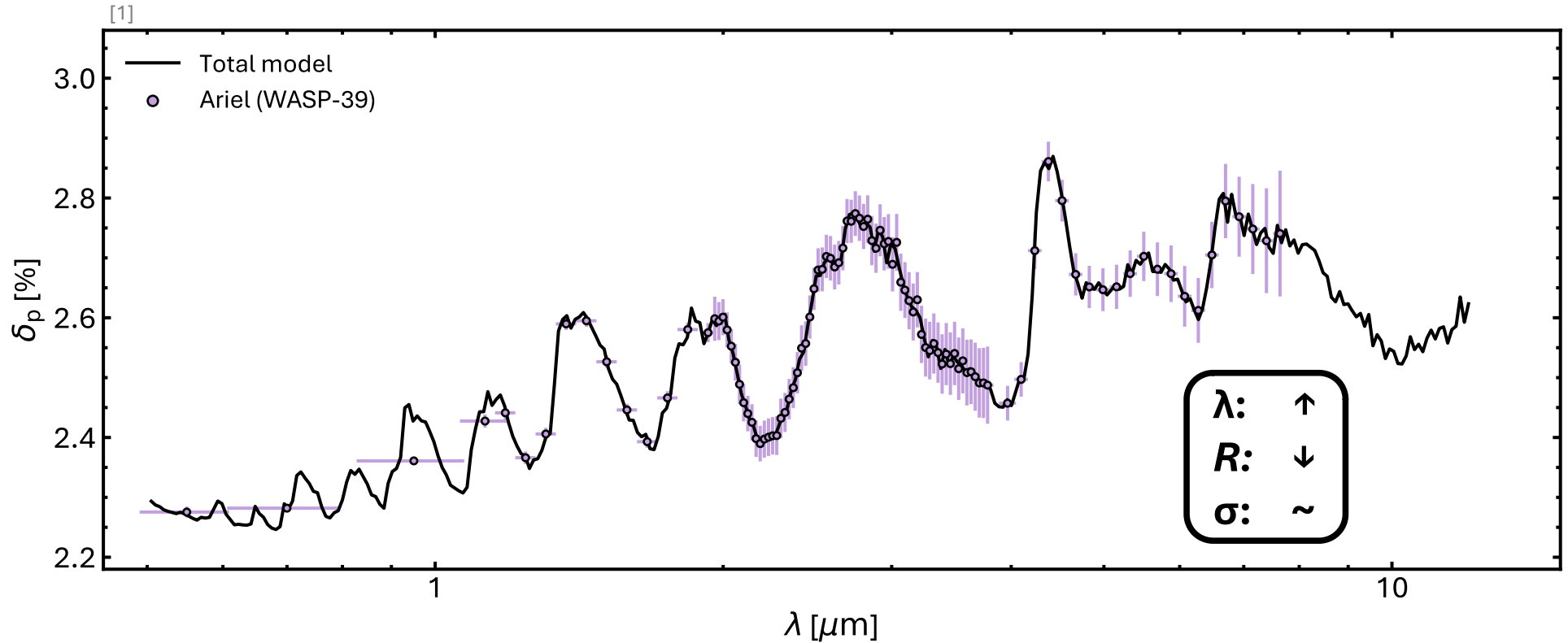
EXAMPLE SPECTRA

Sample provides a mixture of coverage (λ), resolution (R) and precision (σ)



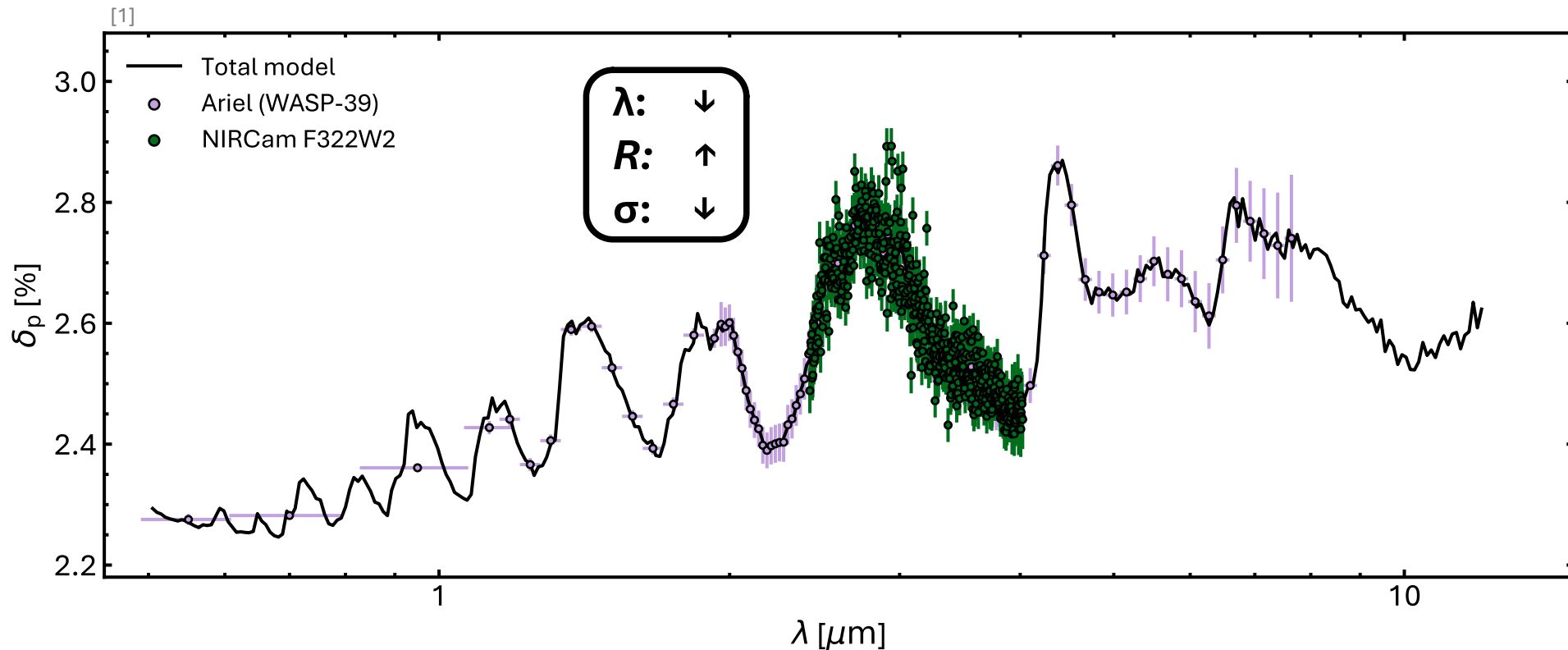
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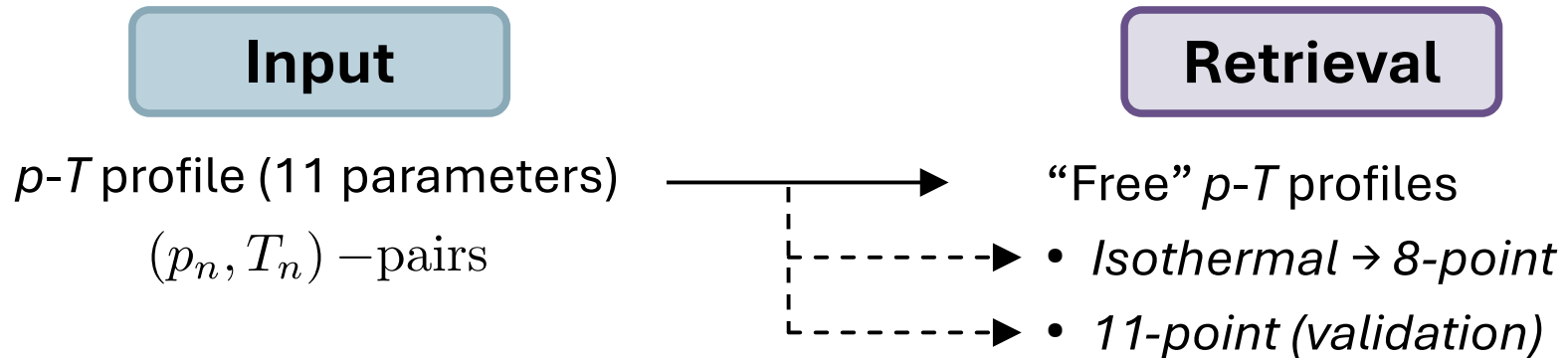


HOMOGENISED RETRIEVALS

- Self-retrieval of R_p , $\log_{10}(X_M)$, $\log_{10}(p_{\text{cloud}})$
 - “Get out what I put in”
 - *How do the biases shown in S24 (simulations for NIRSpec PRISM) translate to different instruments?*

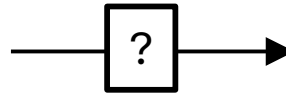
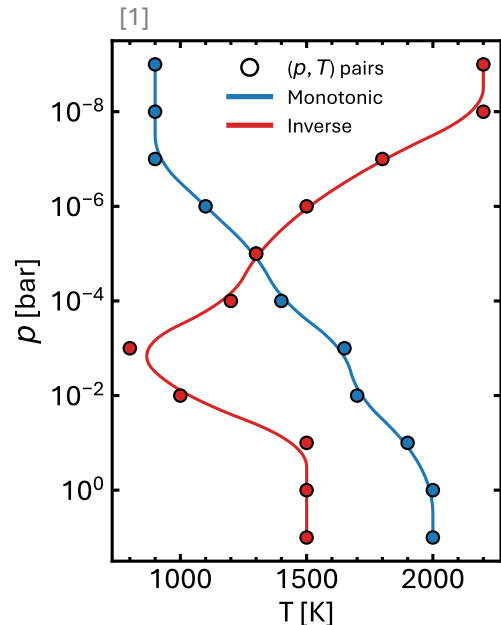
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Supported / necessary
degree of complexity?

Influence of instrument?
(λ -coverage, SNR, ...)

HOMOGENISED RETRIEVALS

- Self-retrieval of R_p , $\log_{10}(X_M)$, $\log_{10}(p_{\text{cloud}})$
- Complexity variation for heuristic p - T profiles
- Evaluate using:

Bayes' factor

Trade-off between fit quality and d.o.f.

Retrieval accuracy

Recovering known input parameters ($\text{CCI}_{99.7}$)

RESULTS

A sample of instrument cases

Bayes' factor

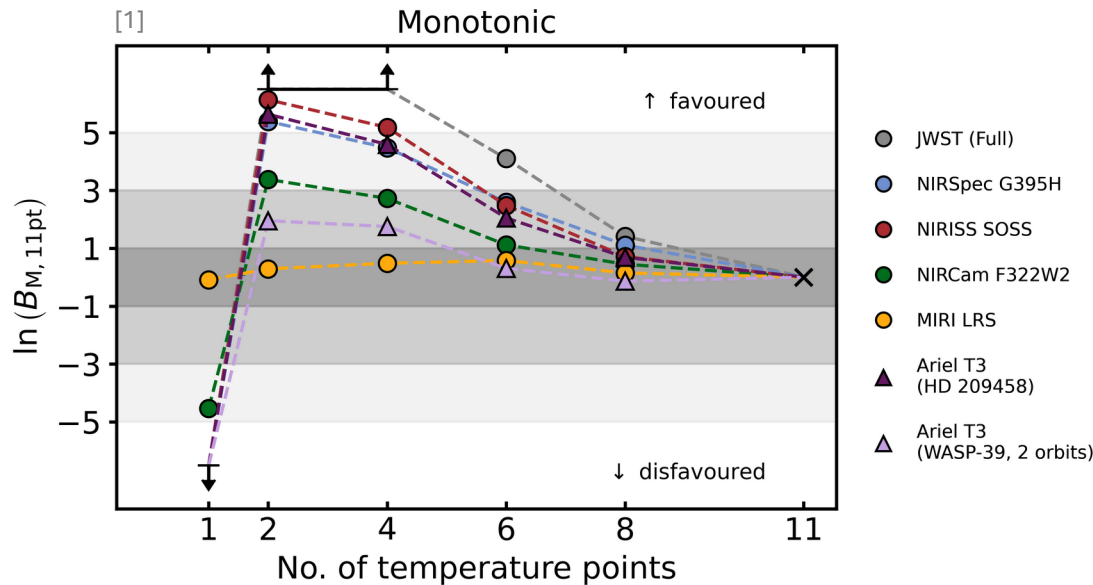
- Variable preference of low-complexity profiles

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Bayes' factor

- Variable preference of low-complexity profiles

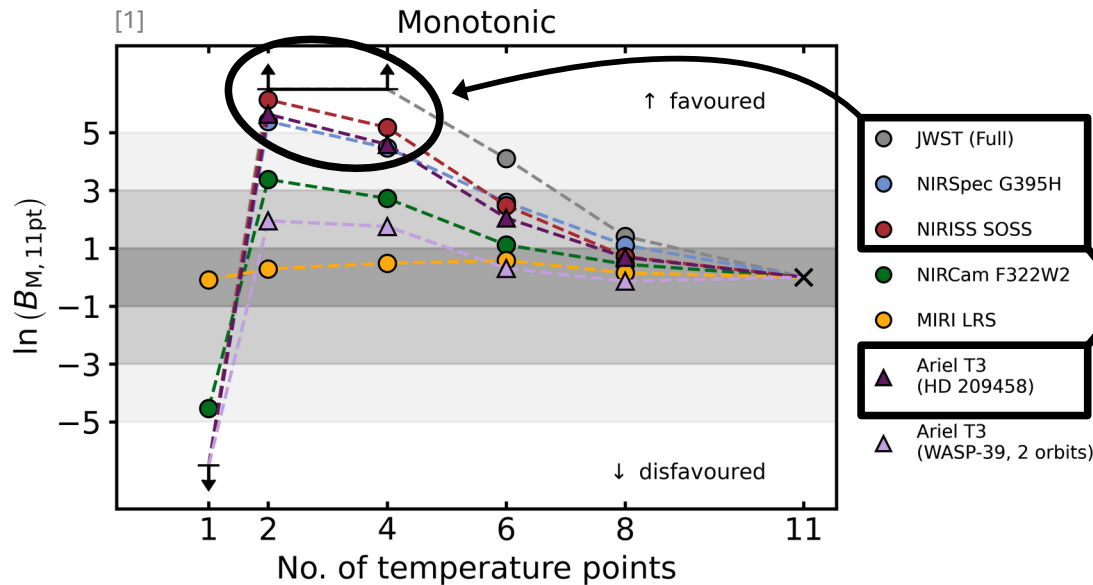


RESULTS

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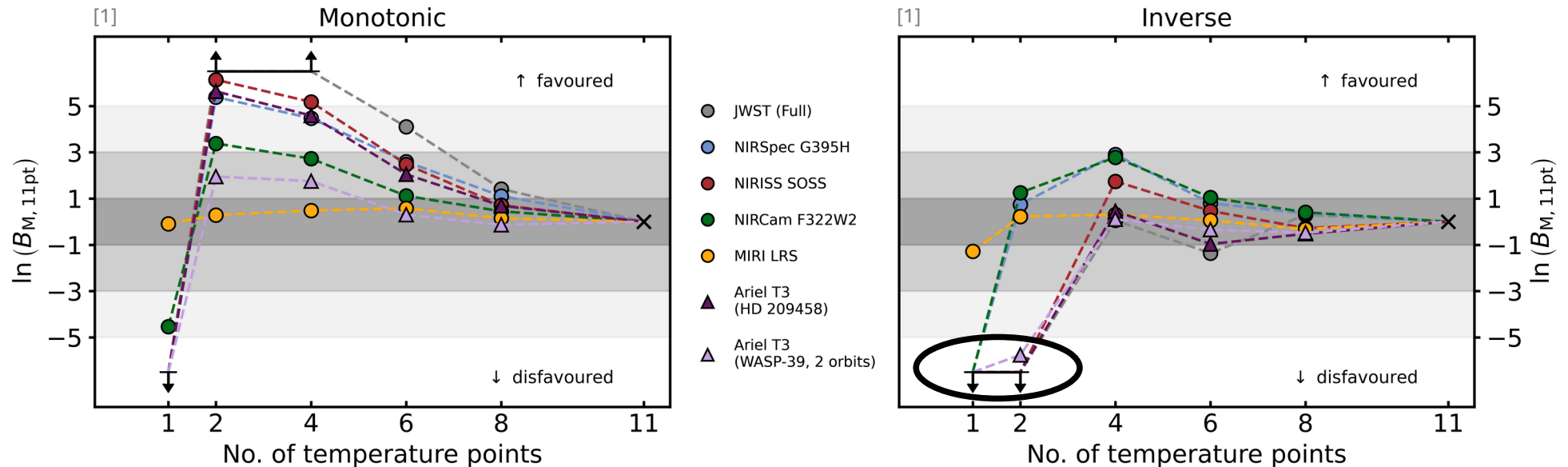
*Magnitude correlated
with λ -coverage*

RESULTS

A sample of instrument cases

Bayes' factor

- Variable preference of low-complexity profiles
 - *Rejection of two-point profile retrievals*



RESULTS

A sample of instrument cases

Retrieval accuracy

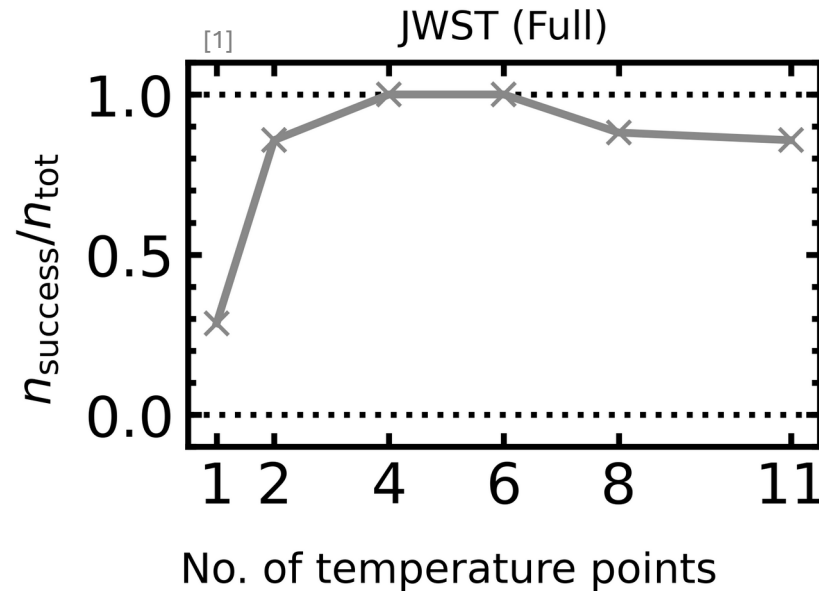
- Accuracy is persistently low in the iso. case

RESULTS

A sample of instrument cases

Retrieval accuracy

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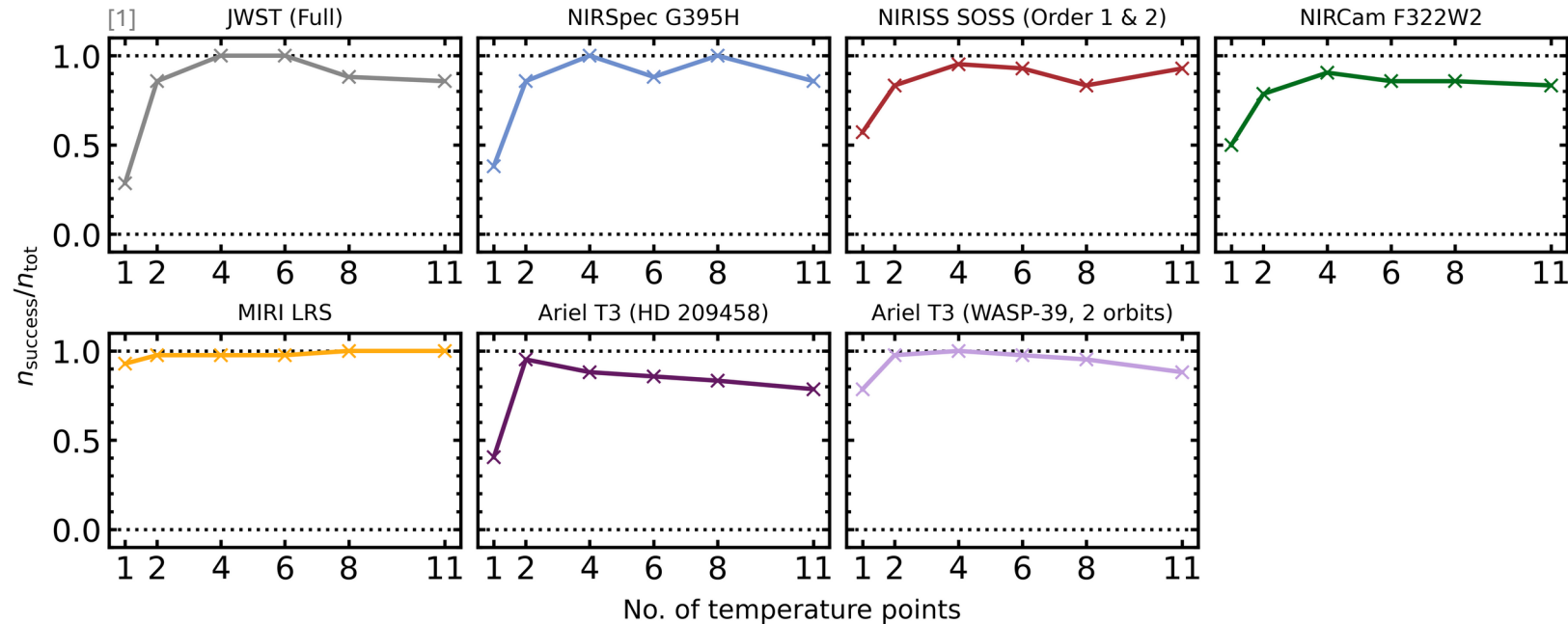


RESULTS

A sample of instrument cases

Retrieval accuracy

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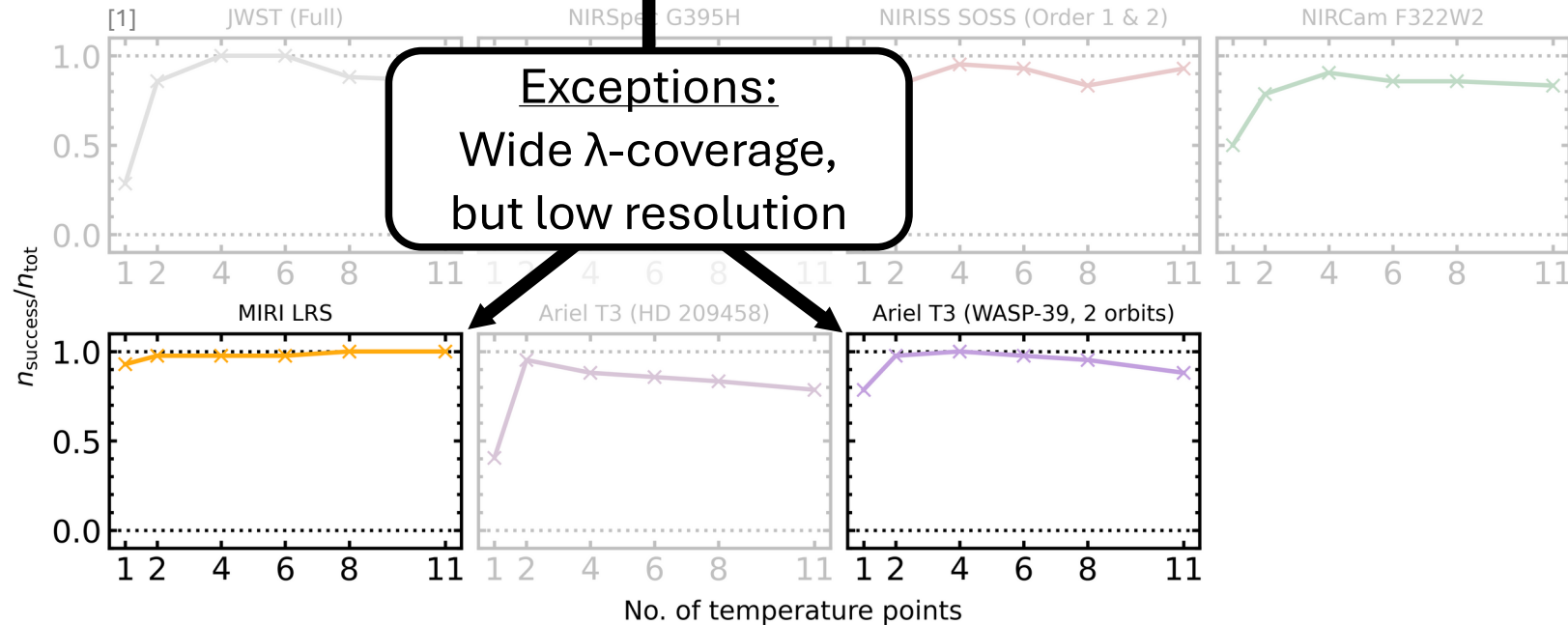


RESULTS

A sample of instrument cases

Retrieval accuracy

- Accuracy is persistently low in the iso. case



CONCLUSION

- Pressure-temperature profiles that are too simple can **bias atmospheric retrieval** results
 - *Degeneracy between retrieved temperature and chemistry (S24)*

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- Pressure-temperature profiles that are too simple can **bias atmospheric retrieval** results
- Isothermal p - T prescriptions **should not be used** in retrievals of state-of-the-art transmission spectra (JWST, Ariel)
 - *Largest impact: Resolution $\rightarrow \lambda$ -coverage*

CONCLUSION

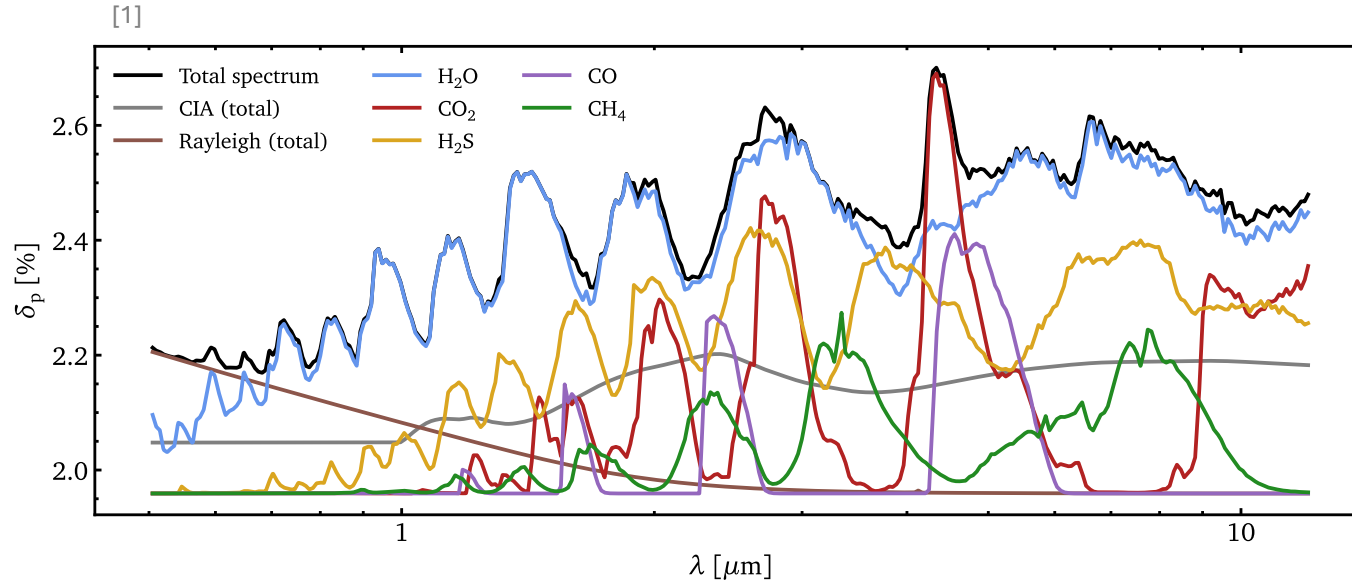
- Pressure-temperature profiles that are too simple can **bias atmospheric retrieval** results
- Isothermal p - T prescriptions **should not be used** in retrievals of state-of-the-art transmission spectra (JWST, Ariel)
- **Simple structure profiles** (2 to 4 nodes) are sufficient to capture atmospheric p - T behaviour
 - *Cloud-coverage mutes these conclusions, as information in the spectra is lost*

Additional Slides

FORWARD MODEL PARAMETERS – SYSTEM

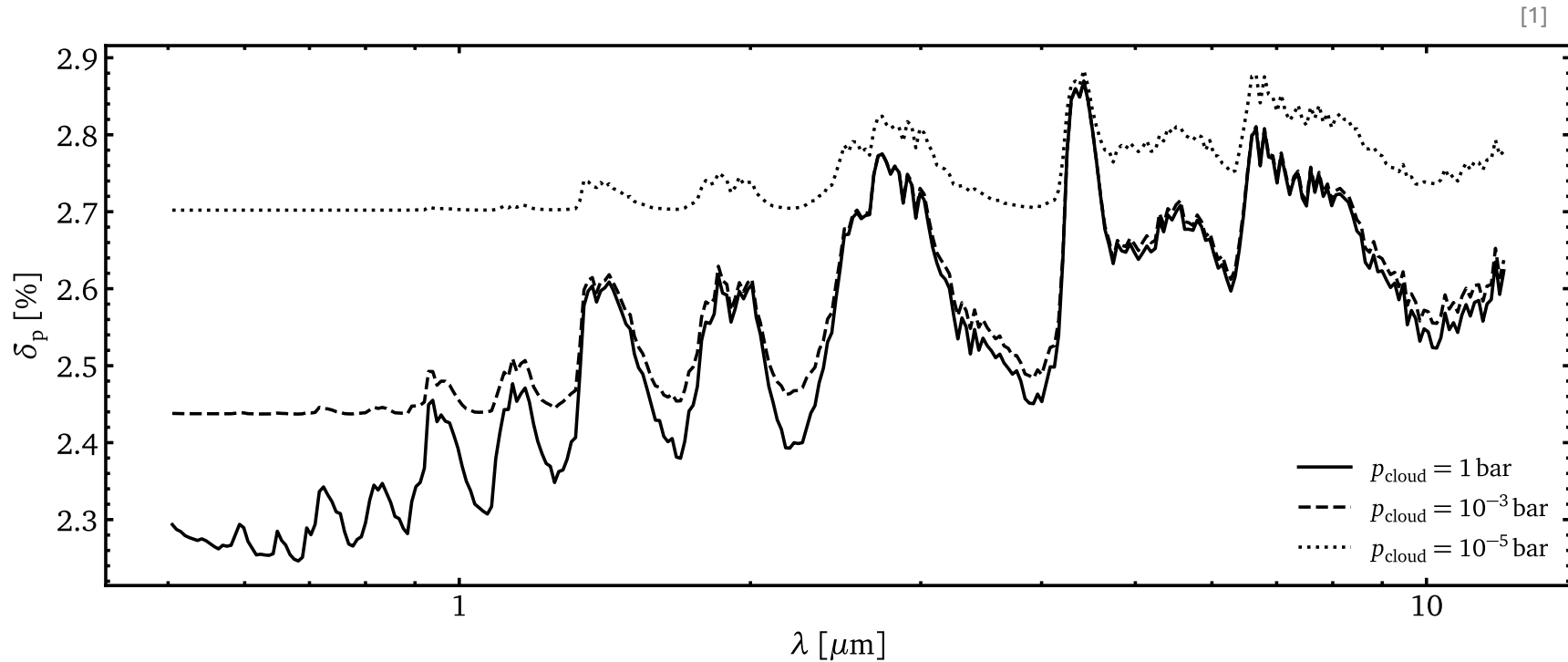
Parameter	Value	Unit
Star		
M_*	0.918 ± 0.047	$M_\odot$
R_*	1.013 ± 0.022	$R_\odot$
T_{eff}	5485 ± 50	K
[Fe/H]	0.01 ± 0.09	dex
$\log g$	4.41 ± 0.15	dex
Planet		
M_p	0.281 ± 0.032	M_j
R_p	1.279 ± 0.040	R_j
P	$4.0552941 \pm 3.4 \times 10^{-6}$	d

FORWARD MODEL PARAMETERS – CHEMISTRY



Parameter	Value
He/H ₂	0.13
$X_{\text{H}_2\text{O}}$	5×10^{-3}
X_{CO}	4×10^{-3}
$X_{\text{H}_2\text{S}}$	5×10^{-4}
X_{CO_2}	5×10^{-5}
X_{CH_4}	1×10^{-7}

FORWARD MODEL PARAMETERS - CLOUDS



INSTRUMENT DETAILS

Instrument	$\lambda_{\min}$ [μm]	$\lambda_{\max}$ [μm]	R^a	σ_{td} [ppm] ^a
JWST				
NIRSpec PRISM	2.25 ^b	5.29	410	104 / 238 ^c
NIRSpec G395H	2.73	5.17	1999	425
NIRISS SOSS	0.63	2.81	834	260
NIRCam F322W2	2.42	4.02	1101	295
MIRI LRS	5.13	11.86	34	239
JWST (full) ^d	0.63	11.86	1391	339
Ariel				
WASP-39, 2 orbits	0.55	7.64	100	369
HD 209458	0.55	7.64	100	100

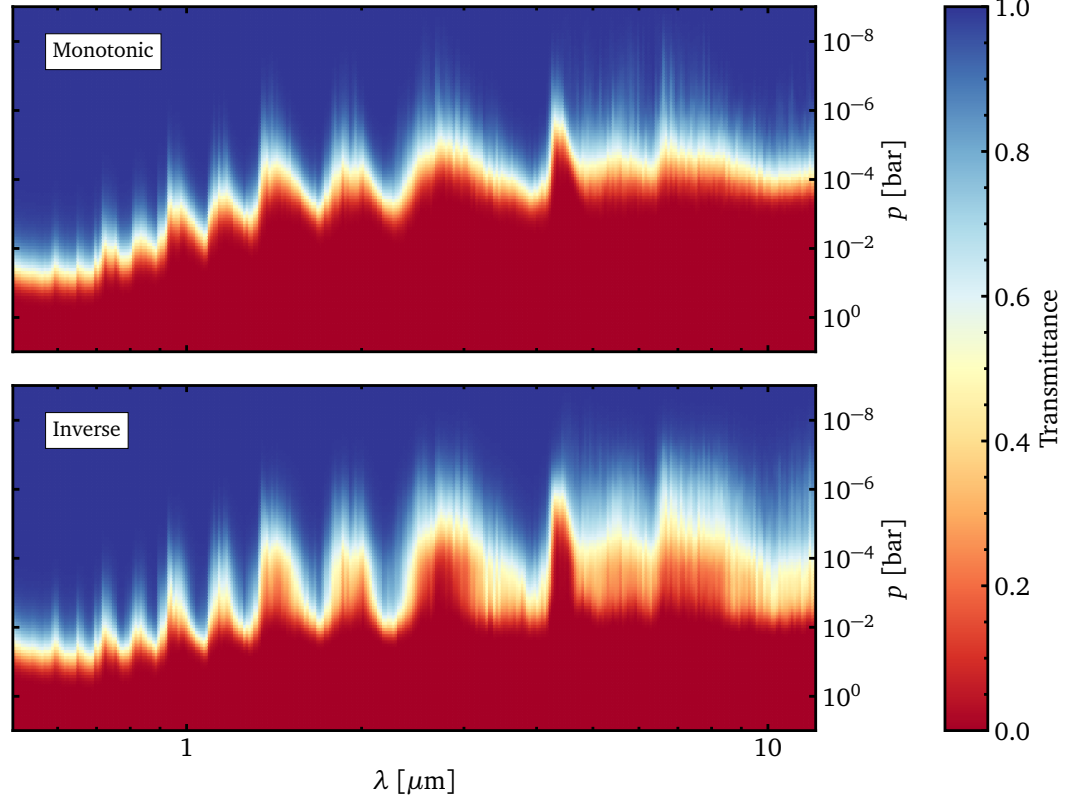
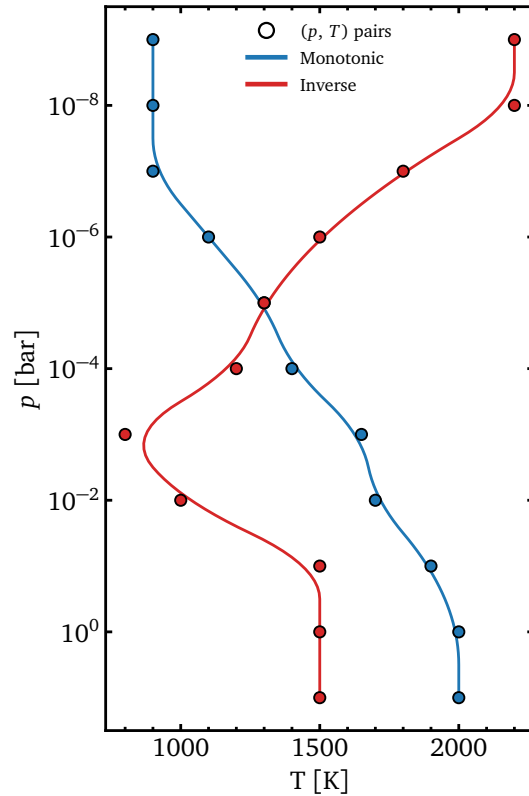
^a The resolution, R , and transit depth uncertainty, σ_{td} , are represented by median values.

^b The NIRSpec PRISM observation is saturated at wavelengths shorter than $\lambda_{\min}$.

^c The first and second value represent the Pandexo and DRS case, respectively.

^d This case uses the noise levels from NIRISS SOSS, NIRSpec G395H and MIRI LRS.

OPTICAL DEPTH INFORMATION

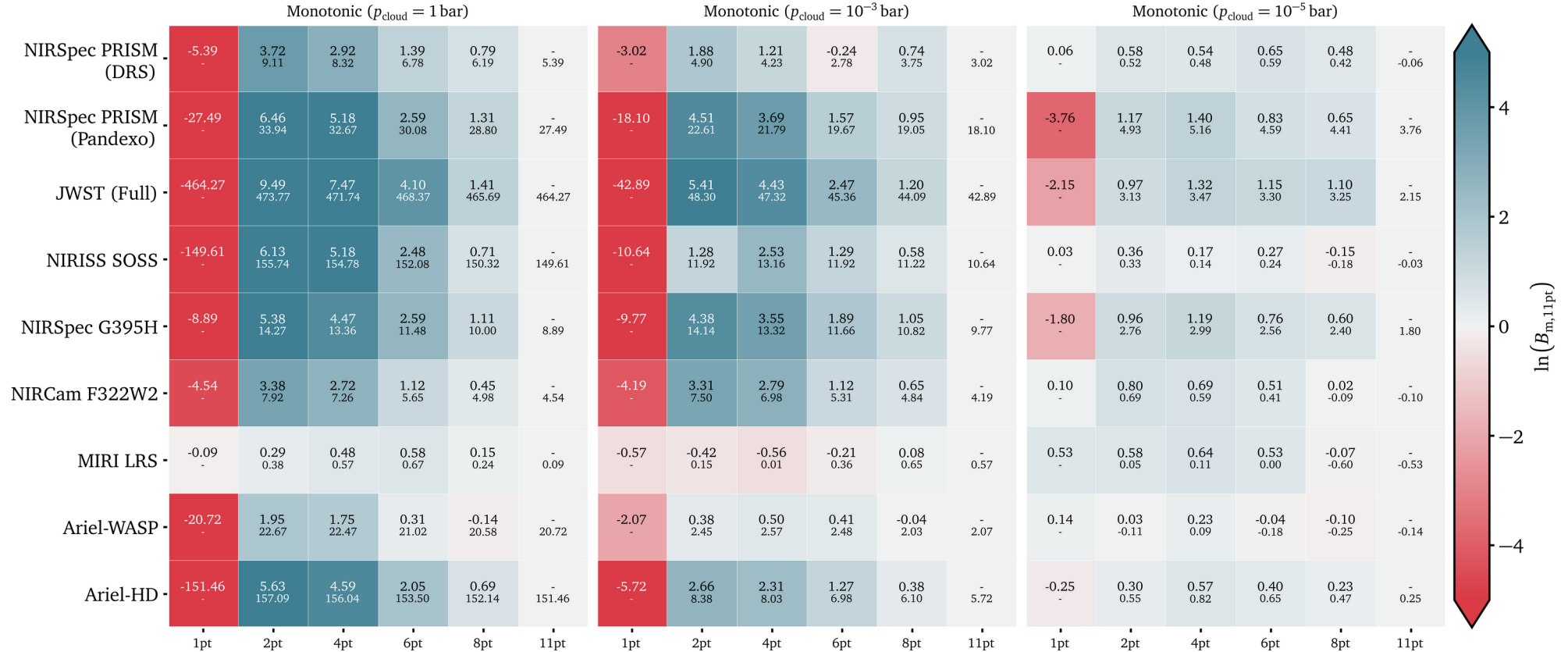


RETRIEVAL PRIORS

Parameter	Prior type	Range	Associated unit
Self-retrieval			
R_p	uniform	$[1.0, 1.5]$	R_J
X_M	log-uniform	$[10^{-12}, 10^{-0.1}]$	–
p_{cloud}	log-uniform	$[10^{-9}; 10]$	bar
Multipoint p–T profile			
T_i	uniform	$[300, 3000]$	K

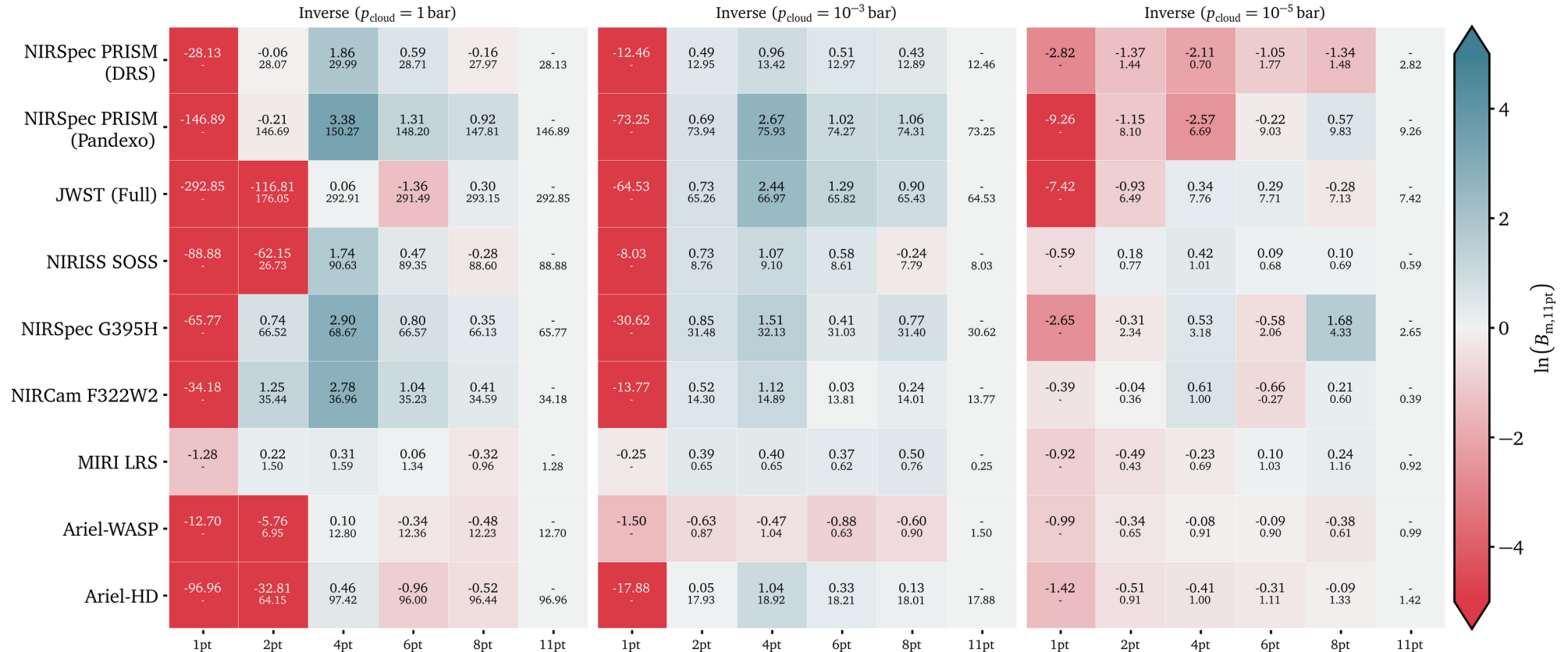
SYNTHETIC STUDY

Bayes factor – “Monotonic” case

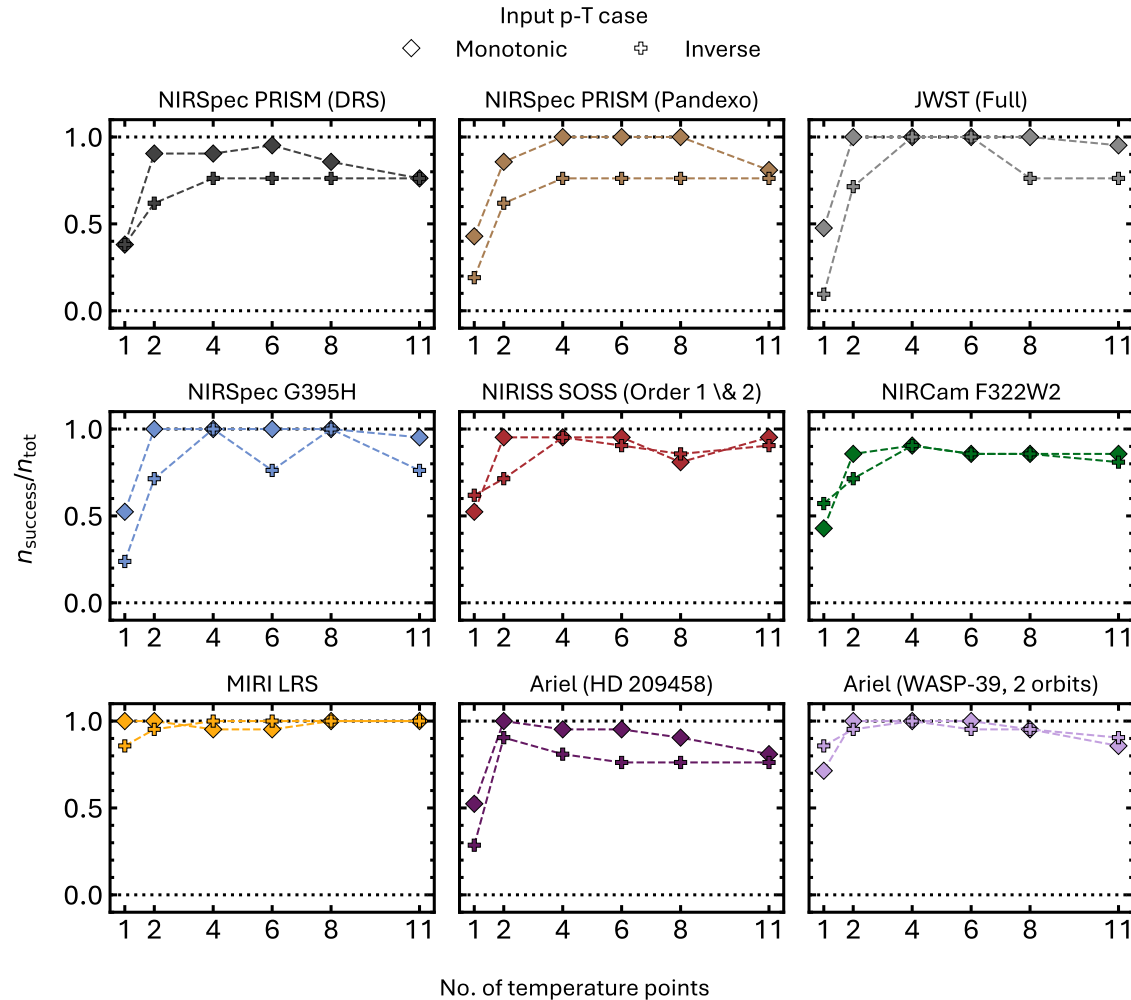


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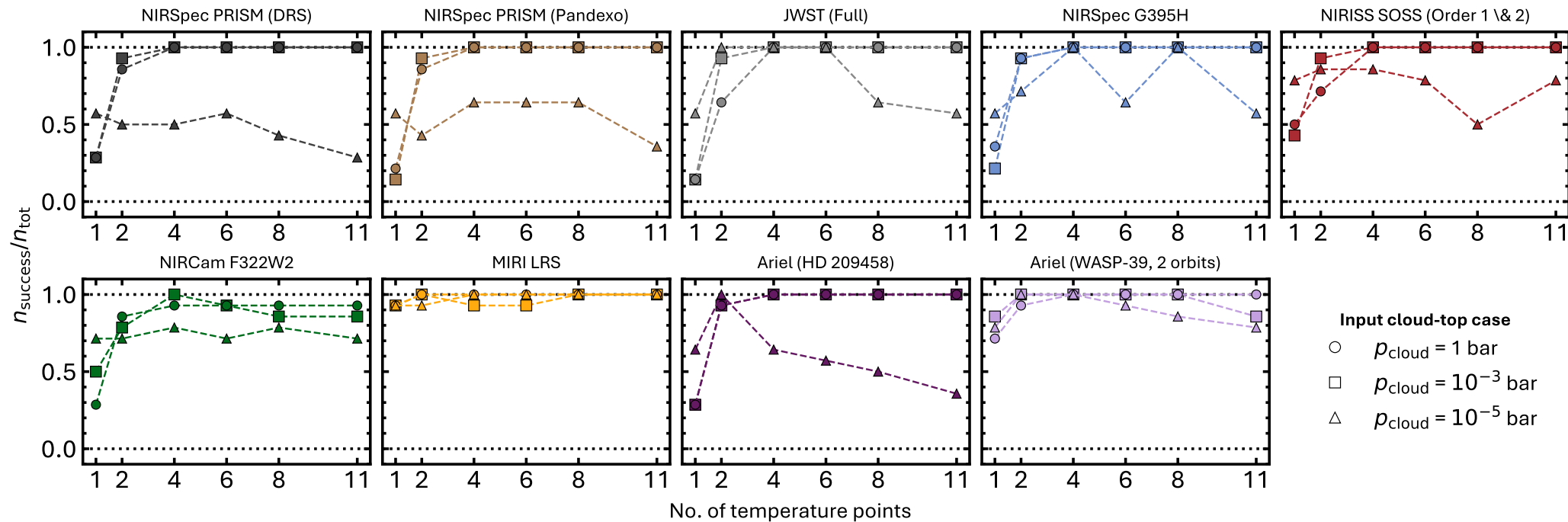
Bayes factor – “Inverse” case



SYNTHETIC STUDY



ACCURACY: CLOUD CASES



REFERENCES

- Al-Refaie, A. F., Changeat, Q., Waldmann, I. P., et al. (2021), ApJ, 917, 37
- Al-Refaie, A. F., Changeat, Q., Venot, O., et al. (2022), ApJ, 932, 123
- Line, M. R., Zhang, X., Vasisht, G., et al. (2012), ApJ, 749, 93
- Blečić, J., Dobbs-Dixon, I., Greene, T. (2017), ApJ, 848.2, 127–151
- Rocchetto, M., Waldmann, I. P., Venot, O., et al. (2016), ApJ, 833.1, 120–133
- Schleich, S., Boro Saikia, S., Changeat, Q., et al. (2024), A&A, 690, A336